# Fetal Health Classification

Machine Learning — Evaluation Project

**Group 16: Junior Nfor & Bakwowi Junior**

TH Deggendorf · Summer Semester 2026

# Motivation

- **Cardiotocography (CTG)** records fetal heart rate and uterine contractions during pregnancy; the standard, low-cost, non-invasive tool obstetricians use to monitor fetal wellbeing in real time.
- **Problem:** correctly separating Normal from Suspect and Pathological patterns matters clinically; missed distress signals can lead to preventable stillbirth or birth injury, while false alarms drive unnecessary interventions such as emergency C-sections.
- CTG traces are still interpreted manually today, and inter-clinician agreement is known to be limited; creating a natural opening for a consistent, data-driven decision-support classifier.
- **Goal:** given 21 diagnostic features automatically extracted from a CTG recording, predict whether the fetal state is Normal, Suspect, or Pathological.
- **Challenge:** the dataset is imbalanced; most recordings are Normal, making standard accuracy a misleading metric.

# Data - Overview

**Source:** Kaggle Fetal Health Dataset

**Citation:** Ayres-de-Campos, D., Bernardes, J., Garrido, A., Marques-de-Sá, J. and Pereira-Leite, L. (2000), Sisporto 2.0: A program for automated analysis of cardiotocograms. J. Matern. Fetal Med., 9: 311-318. https://doi.org/10.1002/1520-6661(200009/10)9:5<311::AID-MFM12>3.0.CO;2-9

**Samples:** 2,126 CTG (Cardiotogram) recordings

**Features:** 21 (signal + histogram statistics)

**Classes:** 3

**Missing values:** 0

**Number of duplicate rows:** 13

**Feature groups:**

- FHR (Fetal Heart Rate) baseline, accelerations, fetal movement, uterine contractions
- Decelerations: light / severe / prolongued
- Short- & long-term variability measures
- FHR (Fetal Heart Rate) histogram statistics (width, min/max, mode, mean, median, variance…)

**Target feature → fetal_health**

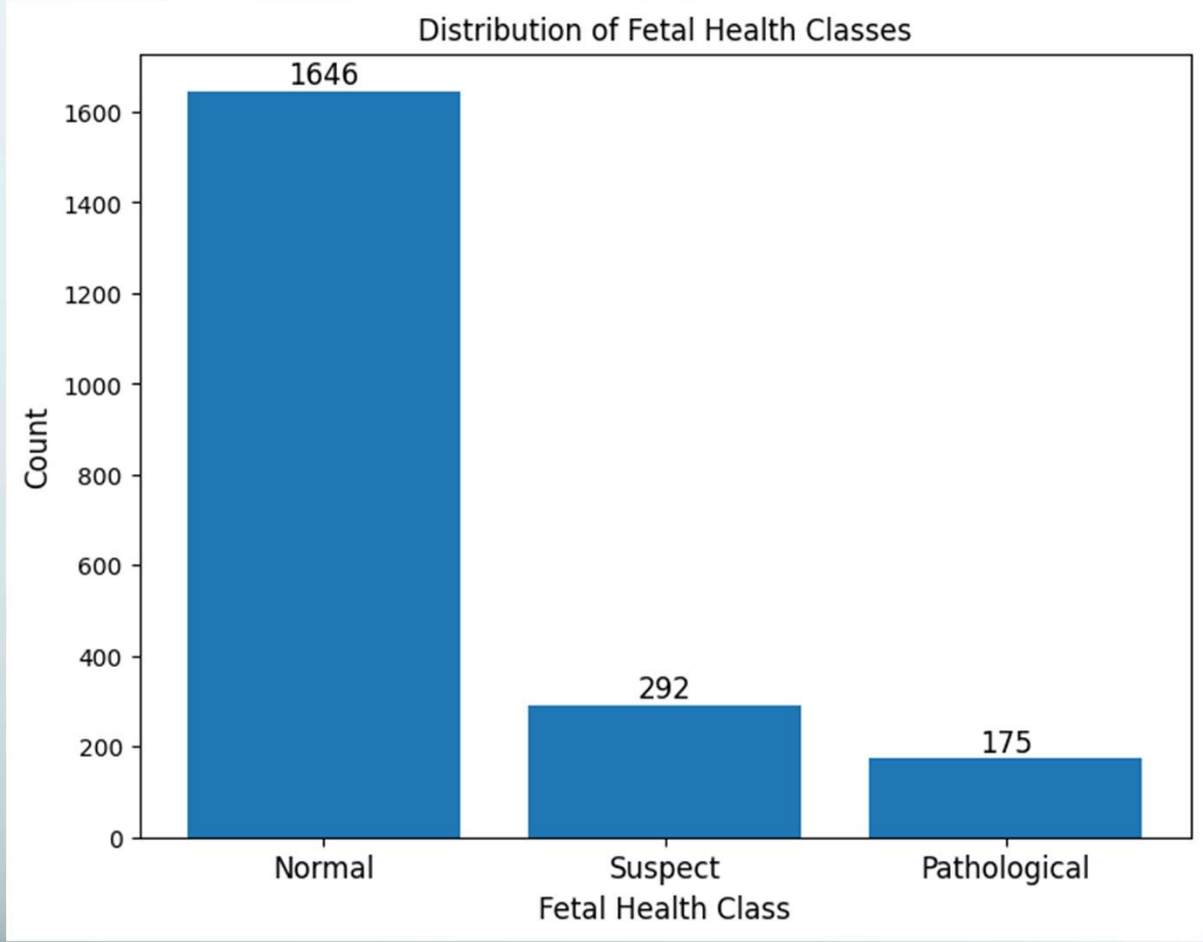
- 1 → Normal
- 2 → Suspect
- 3 → Pathological

# Data - Sample

| baseline value | accelerations | prolongued_decelerations | abnormal_short_term_variability | mean_value_of_short_term_variability | histogram_mean | fetal_health |
|---|---|---|---|---|---|---|
| 120.0 | 0.000 | 0.0 | 73.0 | 0.5 | 137.0 | 2.0 |
| 132.0 | 0.006 | 0.0 | 17.0 | 2.1 | 136.0 | 1.0 |
| 133.0 | 0.003 | 0.0 | 16.0 | 2.1 | 135.0 | 1.0 |
| 134.0 | 0.003 | 0.0 | 16.0 | 2.4 | 134.0 | 1.0 |
| 134.0 | 0.001 | 0.002 | 26.0 | 5.9 | 107.0 | 3.0 |

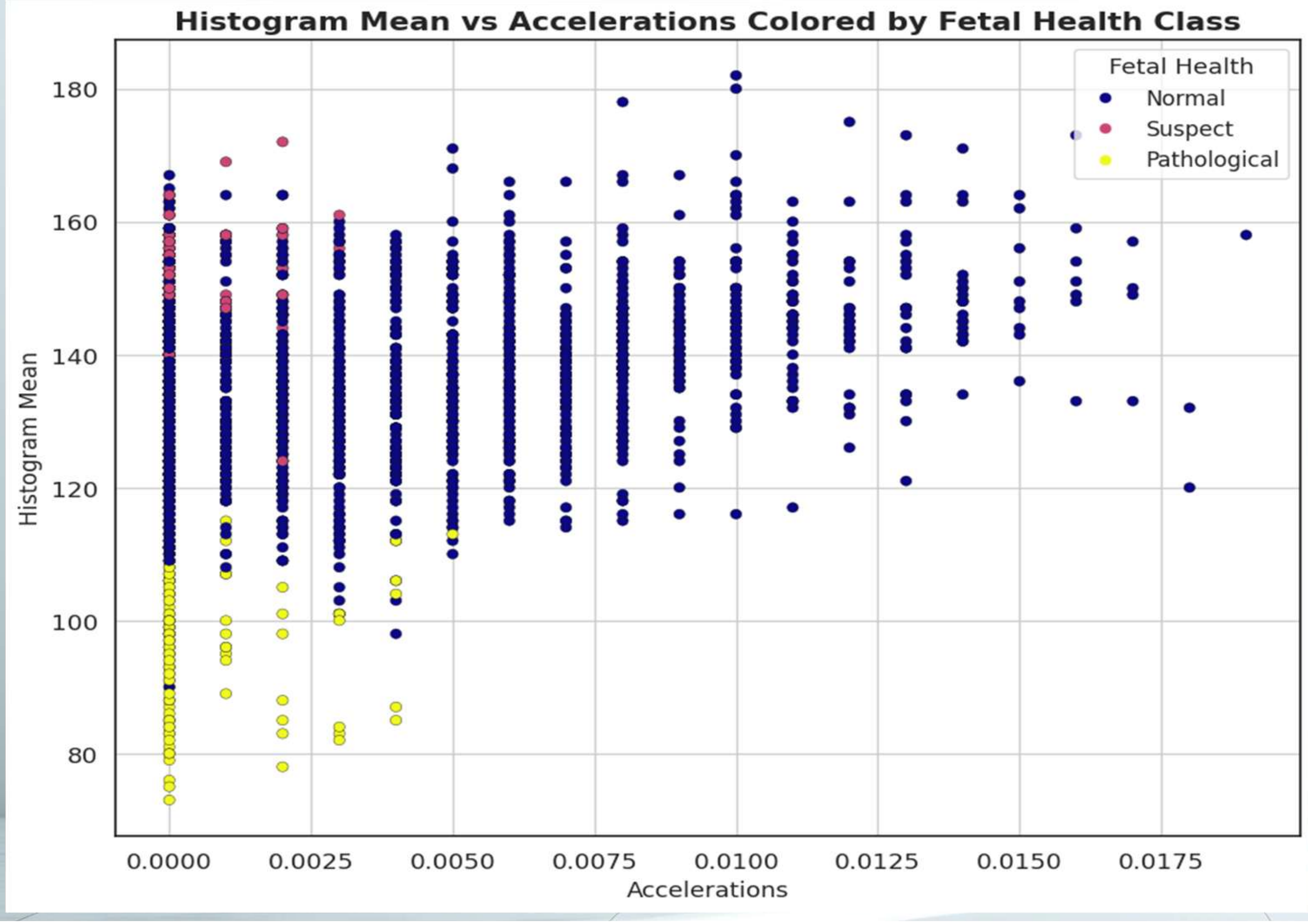
# Data – Class Distribution

- The dataset is significantly imbalanced. Most recordings are Normal.
- Directly motivates macro-F1 as the evaluation metric. Accuracy alone is misleading.
- A stratified train/test split ensures both sets preserve the original class proportions.
- Oversampling minority classes was assessed to counteract the heavy dominant class bias.



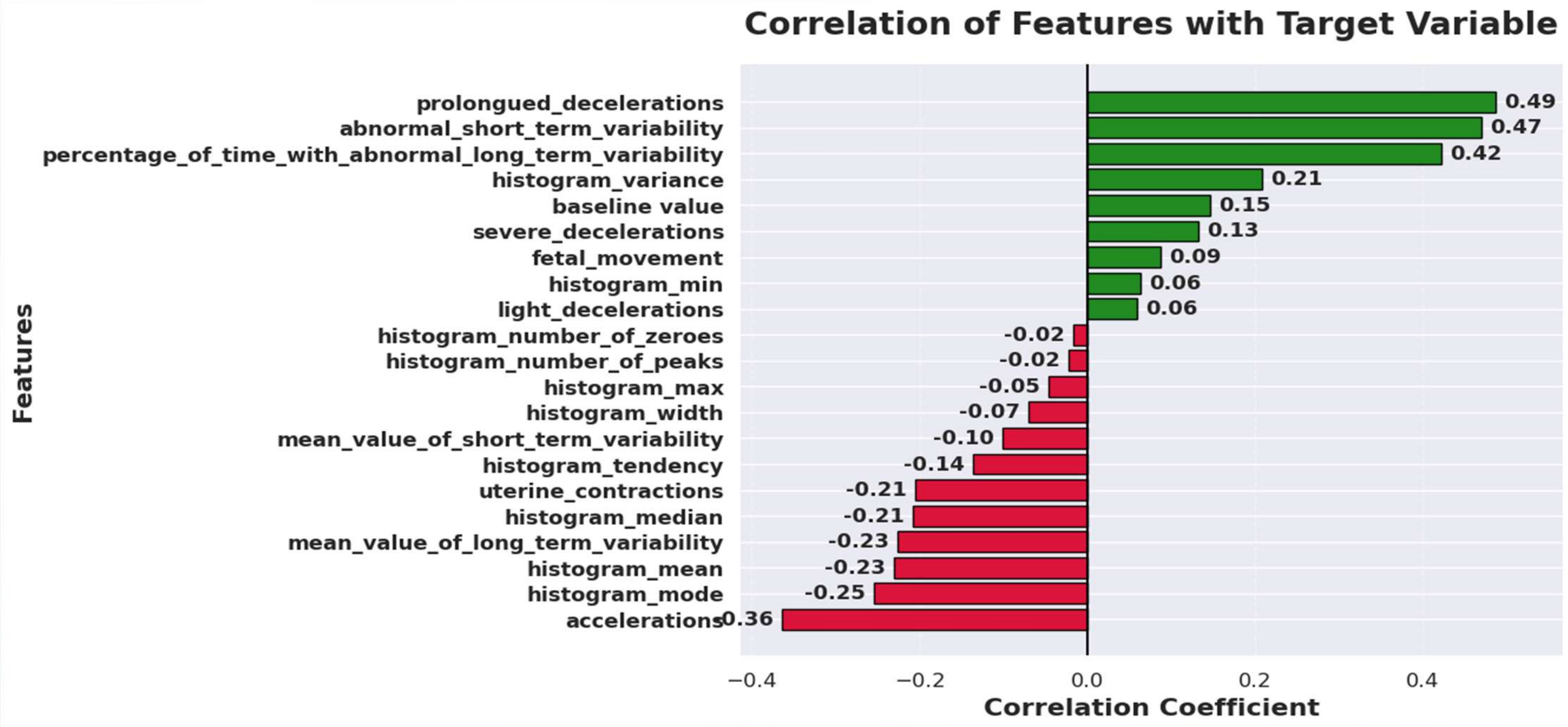
# Data – Feature Exploration and Class Separability

- Pathological cases are clearly separated in most features.
- The Suspect class overlaps heavily with Normal in several features; The main source of classification difficulty.

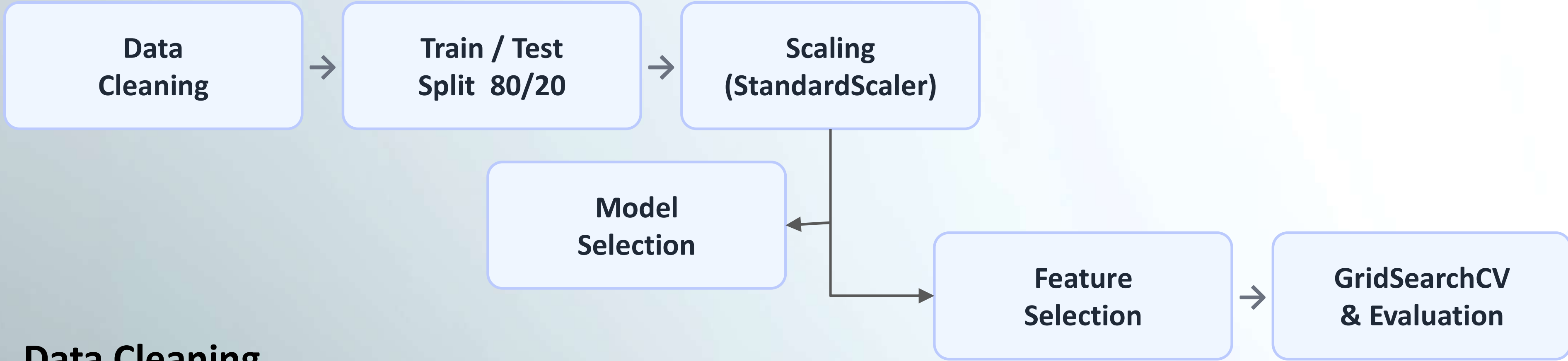


# Data – Feature-Target Correlations

several features have low correlation → motivates feature selection



# Method – Pipeline Architecture

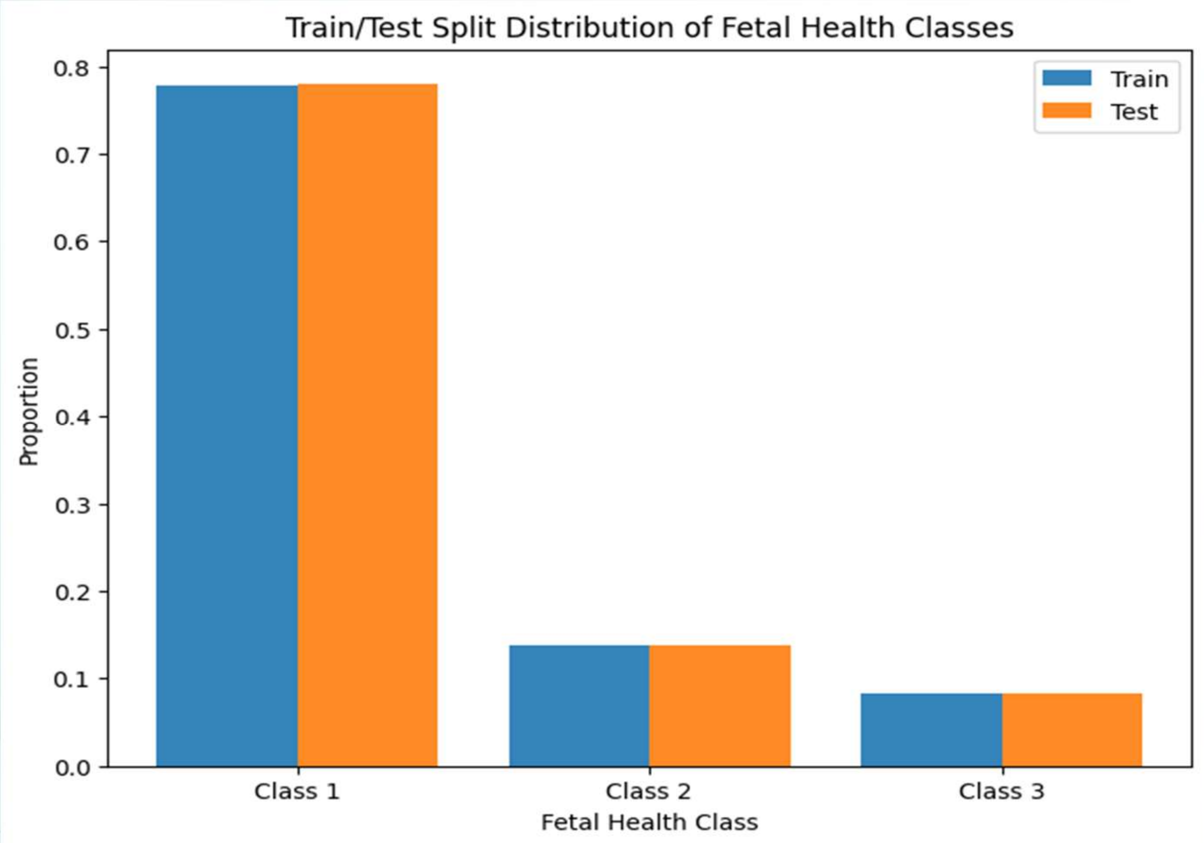



**Data Cleaning**

- 13 duplicates removed (Now 2113 samples).
- No features with missing values.
- No removal of outliers

# Method – Stratified Train/Test Split

- Split ratio: 80% training (1690 samples) · 20% test (423 samples).
- Stratified by target class. Class proportions are preserved in both sets.
- Test set is touched exactly once which is after all tuning is complete.



# Method – Classifiers Selected

## Base Classifiers

**KNN (KNearestNeighbours):**
**Tunned Hyperparameter:** n_neighbours

**Logistic Regression (LogReg):**
**Tunned Hyperparameter:**
- C (Regularization strength)
- l1_ratio

## Advanced Classifiers

**Support Vector Classifier (SVC):**
**Tunned Hyperparameters:**
- C → [0.1, 1, 10, 100]
- Gamma → ["Scale", "auto", 0.01, 1]
- Kernel → ["rbf", "linear", "poly", "sigmoid"]

**Random Forest Classifier (RFC):**
**Tunned Hyperparameters:**
- n_estimators → [100, 200, 300]
- max_depth → [None, 5, 10, 20]
- min_sample_split → [2, 5, 10]
- max_features →["sqrt", "log2"]

# Method – Feature Selection

- Sequential Feature Selector (SFS), forward direction. Adds the single best feature at each step.
- Fixed classifier inside SFS (fixed params). Avoids the cost of nested hyperparameter tuning.
- Grid searches over n_features_to_select: [5, 10, 15, 20].
- RFC uses RFECV instead which leverages the tree's built-in feature importance scores directly.

# Method – Cross-Validation & Hyperparameter Tunning

- GridSearchCV with StratifiedKFold (5-fold outer CV).
- SFS uses 3-fold inner CV. Kept lower to limit computation time.
- **Scoring metric:** macro-F1. Treats all three classes equally regardless of size.
- **Results:** CV mean and std on training data, final score on held-out test set.

**Why macro-F1?**
With 1,646 Normal vs 175 Pathological samples, accuracy is misleading. Macro-F1 treats all three classes equally.

### Handling Class Imbalance

Both **SMOTE (Synthetic Minority Oversampling Technique)** and **class_weight="balanced"** were evaluated. SMOTE consistently outperformed class weighting across all models.

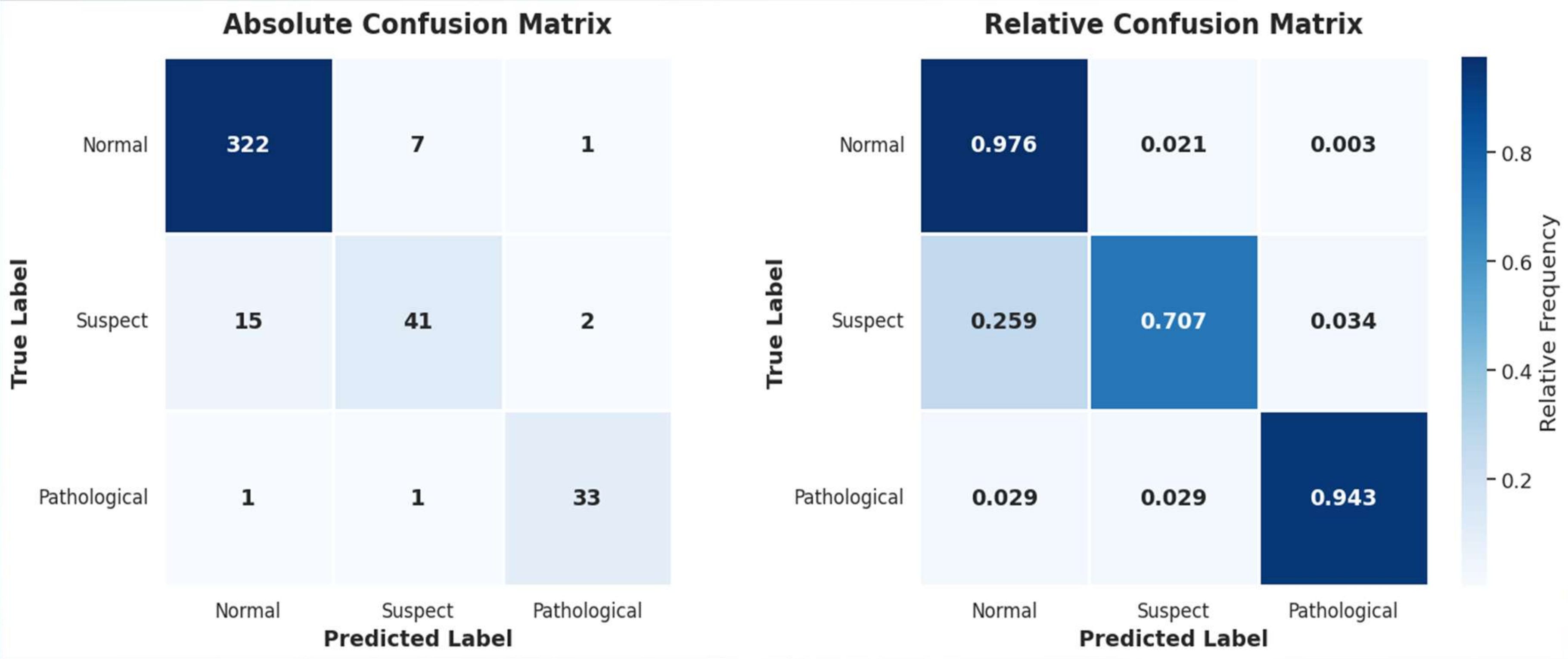
# Results – Model Comparison

All models trained with the same pipeline: scaling → feature selection → GridSearchCV (5-fold, macro-F1).

| Model | Best Params | #_Selected Features | CV F1 (mean, std) | Test F1 (macro) | Test Accuracy |
|---|---|---|---|---|---|
| KNN | k=3 | 10 | 0.8588, 0.0237 | 0.8676 | 0.9338 |
| Logistic Regression | C=10, l1_ratio=0 | 20 | 0.7091, 0.0259 | 0.7804 | 0.8723 |
| SVC | C=100, gamma=0.1, Kernel="rbf" | 20 | 0.8431, 0.0293 | 0.8830 | 0.9385 |
| RFC | Max_depth=None, max_features="sqrt, min_samples_split=2, n_estimators=100 | 18 | 0.8963, 0.0201 | 0.8867 | 0.9362 |

# Results – Best Model

- Confusion matrix of RFC on the held-out test set.
- **Key observation:** Most errors involve the Suspect class; confused with both Normal and Pathological.



# Results – Why RFC outperformed the others

- **Non-linear boundaries:** The Normal/Suspect boundary is non-linear; RFC models it naturally through recursive splits without shape assumptions.

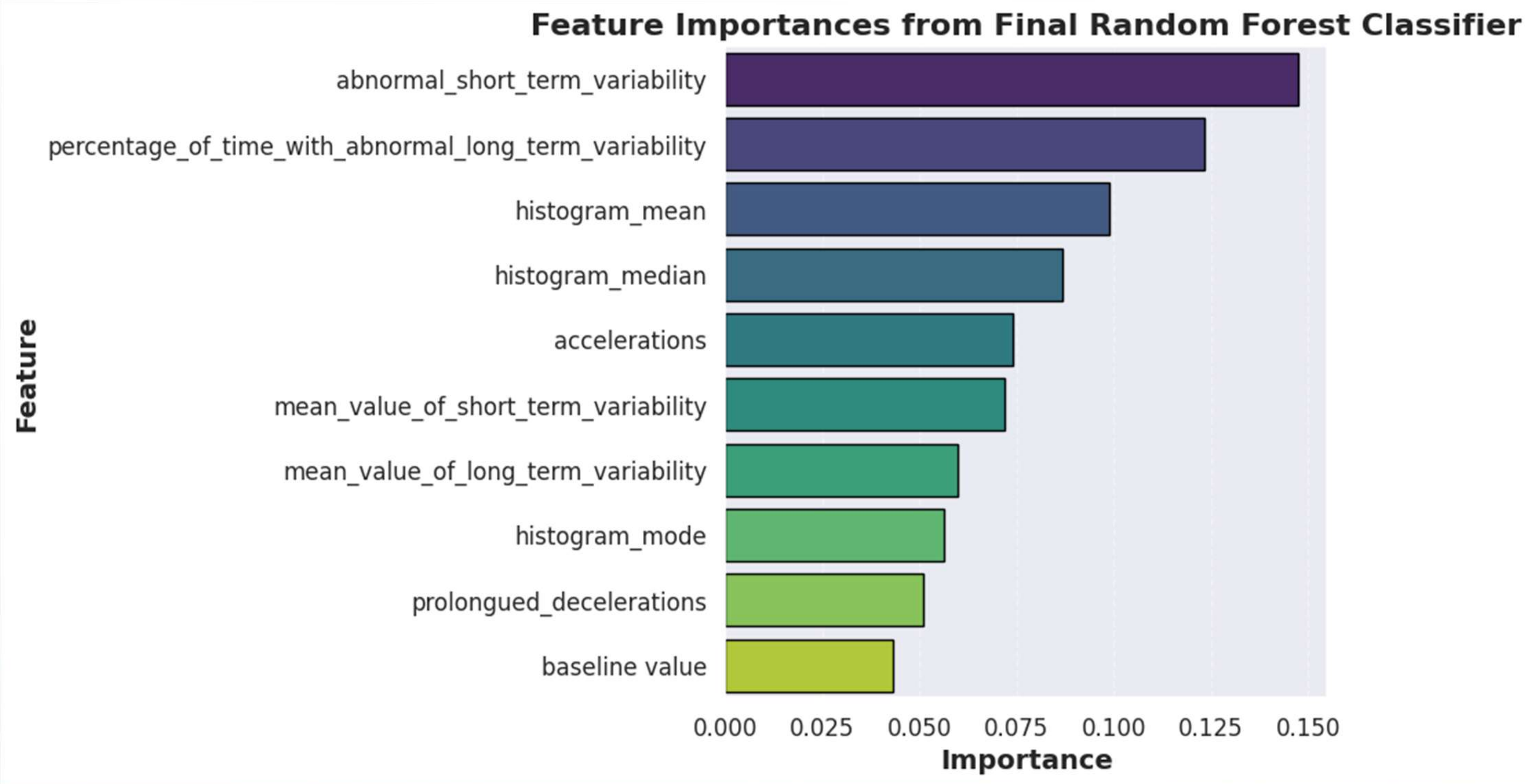
- **Feature interactions:** RFC captures combinations feature at every split; LogReg treats each feature independently.

- **Ensemble stability:** Averaging across hundreds of trees cancels individual errors.

- **No distributional assumptions:** Skewed, non-Gaussian features do not affect tree splits; Logistic Regression and SVC often perform better when features have more regular distributions.

- **Outlier robustness:** Splits are threshold-based, not distance-based, so retained outliers do not distort RFC the way they distort KNN and SVC.

# Results – Why RFC outperformed the others

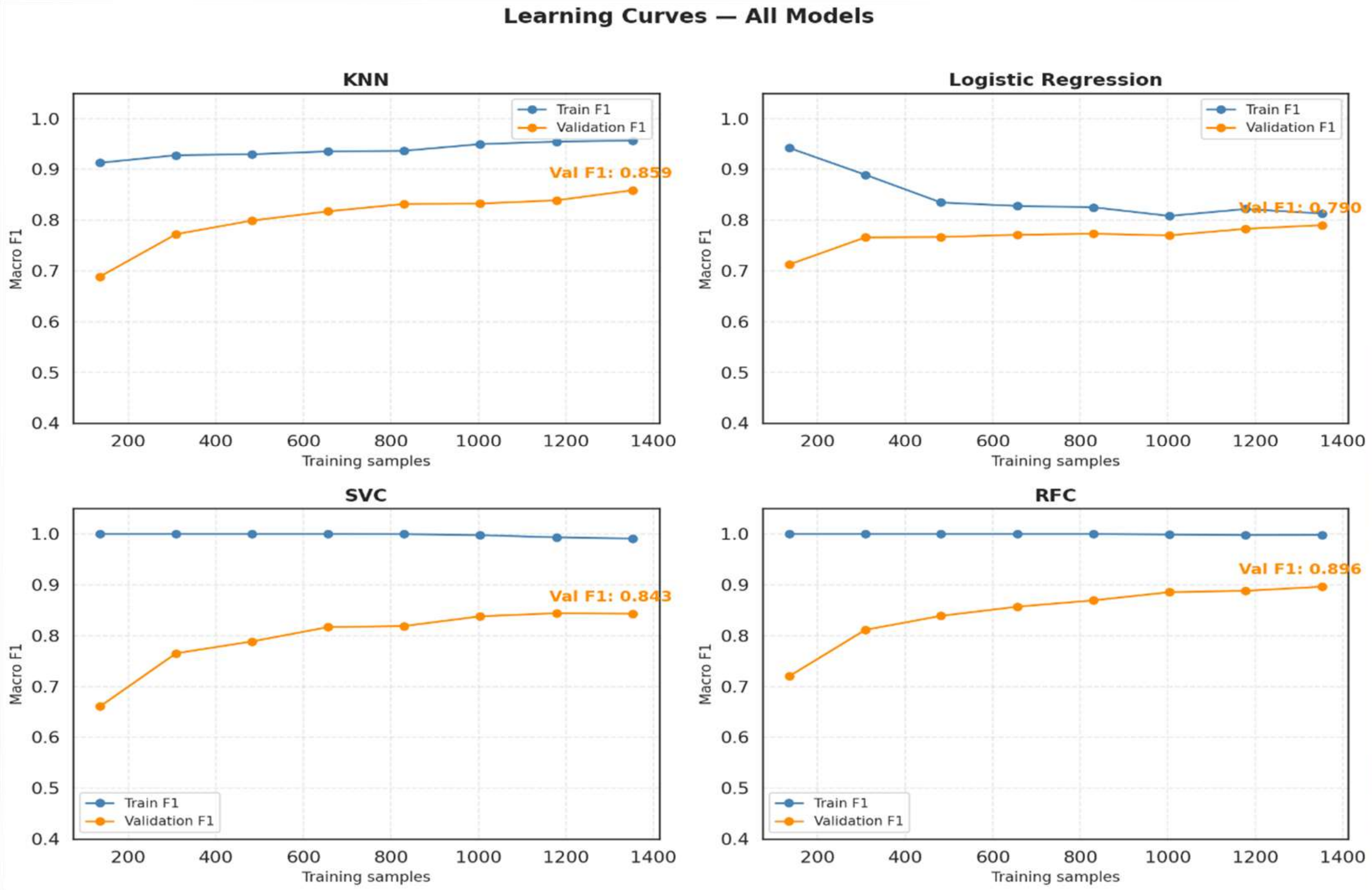


# Discussion

- Advanced classifiers (RFC, SVC) outperformed base classifiers across all metrics confirming that the Normal/Suspect boundary requires non-linear modelling.
- The Suspect class remained the hardest across all four models.
- KNN overfitting (train F1 rising toward 1.0, val F1 = 0.859) at k=1 was the clearest example of how hyperparameter tuning directly impacts generalisation.
- Placing SMOTE inside the CV pipeline was critical; applying it before splitting would have inflated CV scores by leaking synthetic samples into the validation folds.

# Discussion

- The learning curves had not plateaued; more real data could further improve minority class performance, particularly for Pathological.



# Summary

- Dataset of 2,126 CTG recordings, 21 features, 3 imbalanced classes; macro-F1 chosen as the evaluation metric.
- Data cleaning: 13 duplicate rows removed, outliers retained as potential pathological indicators.
- Feature selection reduced 21 features to N most informative; SFS for KNN, LogReg, SVC; RFECV for RFC.
- Advanced classifiers (RFC, SVC) outperformed base classifiers (KNN, LogReg) across all metrics.
- Best model: RFC ; Test F1 = [0.8867] · CV F1 = [mean=0.8963, std=0.0201].
- Main challenge: Suspect class; highest inter-class confusion across all models.

# Thank You – Questions?